# Hand Gesture Recognition, Part 2: A SOTA Pipeline (MediaPipe + MobileNetV2) with an Interactive Demo

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### Abstract

In Assignment 1 I classified three hand gestures (peace, stop, thumbs_up) from a self-collected dataset of 303 images and found a methodological weakness: every class was photographed in a single short session, so a classifier can exploit the background or near-duplicate frames instead of the hand. In Assignment 2 I add a state-of-the-art component, Google's **MediaPipe Hand Landmarker**, which localizes the hand so the classifier sees only the hand region. I compare a baseline MobileNetV2 on the full image against the same network on the MediaPipe crop (clean test accuracy 1.000 vs 0.978 — both near ceiling on this leaky split) and probe **what each model actually uses**. Changing the background leaves both unchanged, but **hiding the hand** drops the baseline to 0.522 (from 1.000) while the crop model, by construction, cannot use the background at all. So the full-image baseline carries a measurable background/context shortcut that the MediaPipe crop removes by design. I add Grad-CAM explanations, an ablation, latency measurements, and an interactive web demo (camera → prediction → Grad-CAM).

## 1. Introduction

Gesture recognition is used in touchless interfaces, sign-language tools and games. In A1 I built the full pipeline (collect, preprocess, train, evaluate) and compared an SVM (82.6%), a small CNN (100.0%) and a fine-tuned MobileNetV2 (97.8%). Two results from A1 motivate this work. First, the simple CNN reached 100% on the test set — implausibly high. Second, I noted in A1 that photos taken one after another look almost identical, so near-duplicates may land in both train and test. This is **data leakage**, and it means the headline accuracy overstates real performance. A2 therefore has two goals: (i) integrate a modern SOTA component that attacks the root cause (background dependence), and (ii) measure performance in a way that a leaky random split cannot hide — a robustness stress test and visual explanations.

## 2. Dataset update

I keep the A1 dataset: 303 JPG images, one folder per class (peace 100, stop 100, thumbs_up 103), all of my own hand, taken indoors. I split it stratified 70/15/15 (212/45/46) with seed 42 — the same protocol as A1 for a fair comparison. **No synthetic or generated media is used; the synthetic share is 0%.**

**Leakage finding (new).** Inspecting the file timestamps shows each class was captured in one burst of about a minute (thumbs_up 12:10–12:11, stop 12:13, peace 12:17). So within a class the frames are near-duplicates and a random split cannot separate sessions. I keep the random split for comparability with A1, but I treat its accuracy as an optimistic upper bound and rely on the stress test (Section 6) for the real signal.

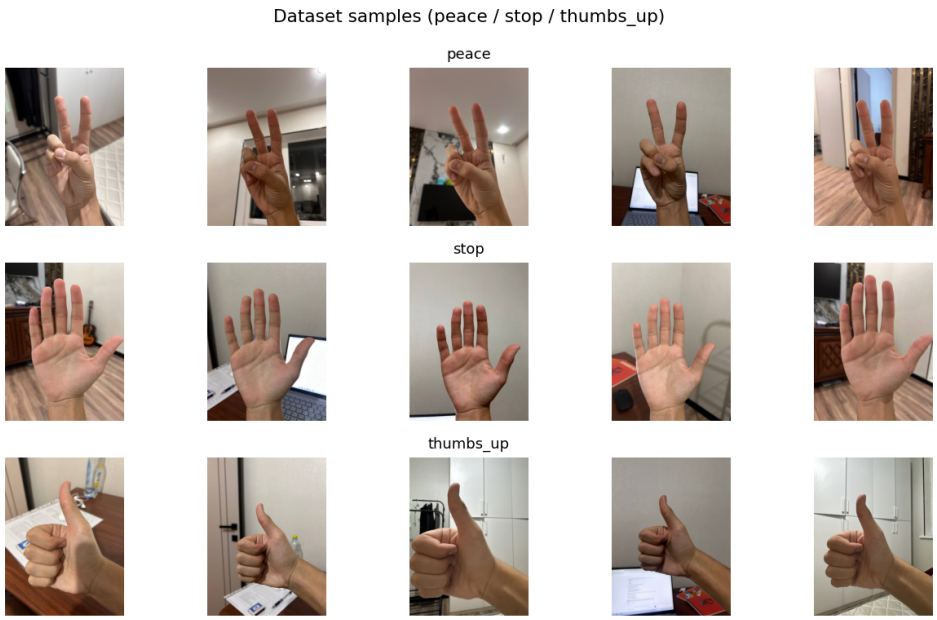



Figure 1. Random samples from each class.

**MediaPipe detection rate:** a hand was detected in 96.0% of images; the few misses fall back to a center crop so the pipeline never fails.

## 3. SOTA method: MediaPipe Hand Landmarker

**What it is.** MediaPipe Hand Landmarker (Google) is a real-time, on-device hand perception model. A palm detector first finds the hand, then a landmark model regresses 21 3D keypoints (wrist, knuckles, fingertips). It is the current production-grade standard for hand tracking and runs in the browser and on phones. I use the modern *mediapipe.tasks* API with the *hand_landmarker.task* model, as the assignment allows choosing a modern SOTA.

**How I use it.** For every image I run the landmarker, take the 21 keypoints, build a square bounding box around them (expanded by 35%), and crop the image to that box before classification. The classifier (MobileNetV2, same as A1) therefore sees mostly the hand, not the room behind it.

**Why it helps this topic.** The A1 weakness is that the model can key on the background. By removing the background before classification, the MediaPipe crop forces the decision onto hand shape, which is the actual signal. It also standardizes the hand's scale and position, and the 21 landmarks are a free, interpretable by-product. This directly realizes the 'crop the hand with a detector such as MediaPipe' improvement I proposed in the A1 future-work section.

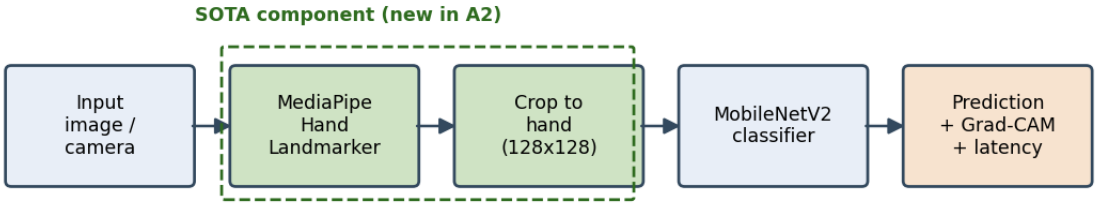



Figure 2. The A2 pipeline. The dashed box is the new SOTA component.

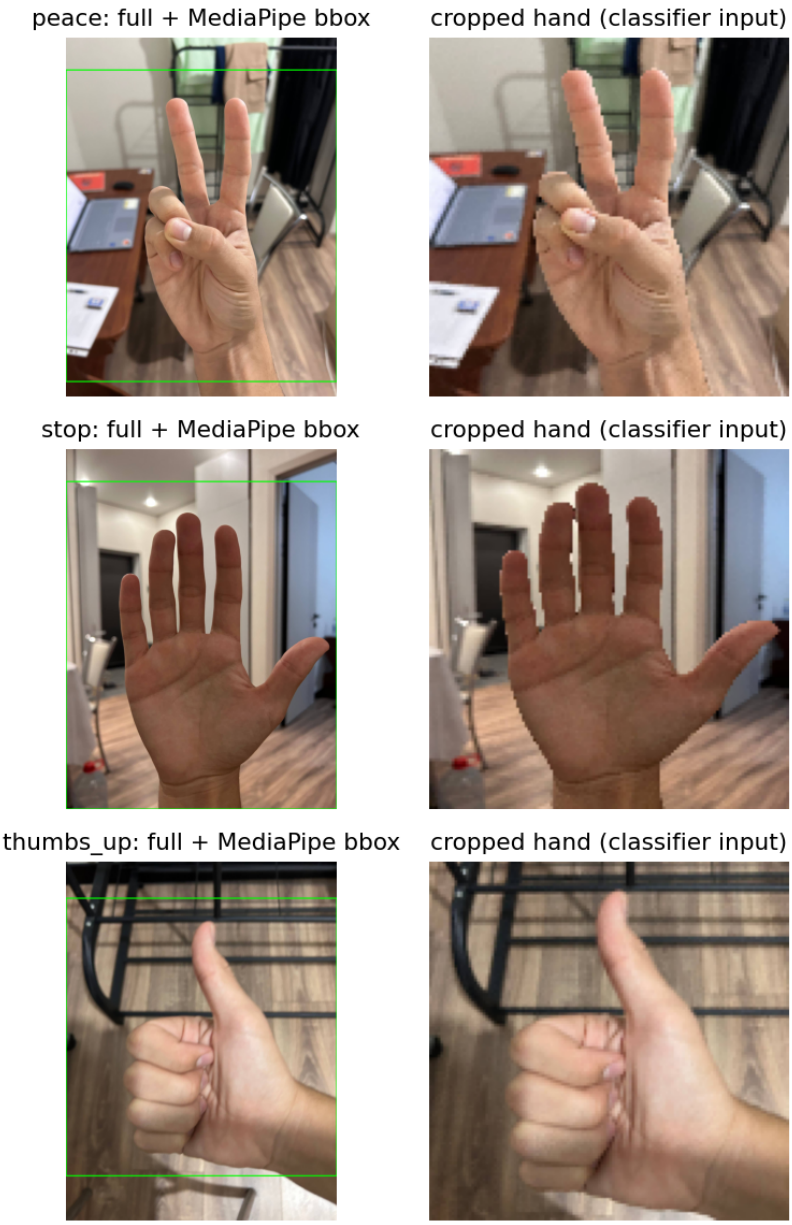



Figure 3. MediaPipe bounding box (left) and the cropped classifier input (right).

**On the assignment's recommended generators.** The hand-gesture row suggests Step-Video-T2V and ArtAug to synthesize motion/lighting variations. These are heavy offline generators; within the time budget and for a defensible, reproducible result I chose a SOTA component that runs live and attacks the identified weakness head-on. Synthetic augmentation is discussed as future work.

## 4. Method

Both models share the architecture and hyper-parameters, so any difference comes from the input, not the tuning. Backbone: MobileNetV2 pretrained on ImageNet, input 128×128 scaled to [-1,1], global average pooling, dropout 0.3, softmax over 3 classes. Training is two-stage: 15 epochs head-only (Adam 1e-3), then fine-tune the last 30 layers (Adam 1e-5), batch size 16, early stopping on val loss. Augmentation (flip, small rotation/zoom/shift, brightness/contrast) is applied to the training images only. **Baseline** = full image; **SOTA** = MediaPipe crop.

## 5. Clean test-set results

| Model | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| A1 MobileNetV2 (full, reported) | 97.8% | - | - | - |
| Baseline MobileNetV2 (full) | 1.000 | 1.000 | 1.000 | 1.000 |

| | | | | |
|---|---|---|---|---|
| SOTA MobileNetV2 (MediaPipe crop) | 0.978 | 0.980 | 0.978 | 0.978 |

Table 1. Clean test set (46 images, macro-averaged). On this leaky split both models score very high — as expected, and exactly why the clean number is not the real story.

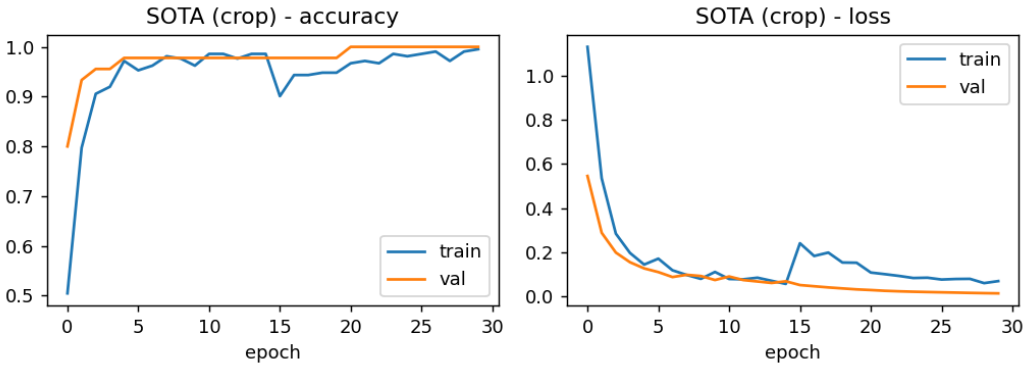



Figure 4. Training/validation curves (SOTA model).

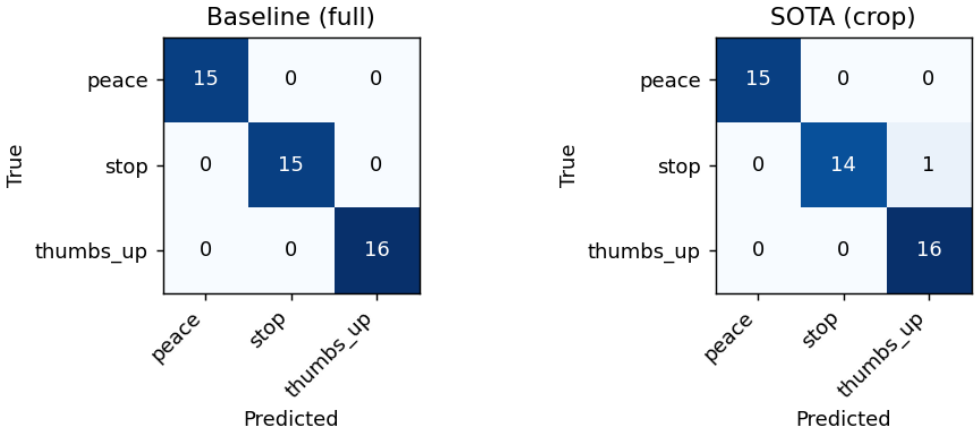



Figure 5. Confusion matrices: baseline (left), SOTA (right).

## 6. What does each model actually use? (robustness probes)

Because the clean split is leaky, high accuracy alone proves nothing. So I probe the *cause* of the predictions with two controlled interventions on the test set, using the MediaPipe hand box to know where the hand is. **(a) Change the background:** replace everything outside the hand box with a random solid color, random noise, or another image's background (plus a strong photometric shift). **(b) Hide the hand:** gray out the hand box and keep the background. If a model needs the hand, (a) should not hurt it and (b) should break it.

| Condition | Baseline (full) | SOTA (crop) |
|---|---|---|
| clean | 1.000 | 0.978 |
| bg_color | 0.978 | 0.978 |
| bg_noise | 1.000 | 0.978 |
| bg_image | 0.978 | 0.978 |
| photometric | 1.000 | 1.000 |
| hand_hidden | 0.522 | 0.326 |

Table 2. Probe accuracy. **Changing the background changes almost nothing** for either model — the hand fills most of these close-up photos, so there is little background to exploit. But **hiding the hand drops the baseline from 1.000 to 0.522** (chance is 0.33): the full-image model relies mostly on the hand, yet still extracts a non-trivial amount from the background/context. The SOTA crop sees *only* the hand, so a blank hand leaves it at chance (its hand_hidden value is a control). In short, the baseline has a measurable

background/context shortcut; the MediaPipe crop removes it by construction.

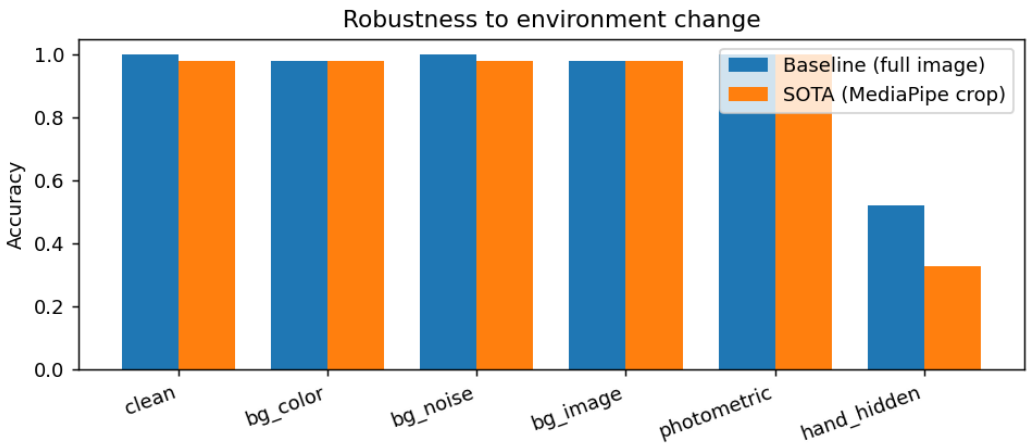



Figure 6. Probe accuracy. bg_* / photometric = background changed; hand_hidden = hand grayed out (for the crop model this is a chance-level control).

### Ablation.

Removing data augmentation from the SOTA model changes clean accuracy to 1.000 (F1 1.000), isolating augmentation's contribution. The main ablation is the SOTA comparison itself: with vs without the MediaPipe crop (Tables 1–2).

## 7. Explainability (Grad-CAM)

Grad-CAM highlights the pixels that drive the predicted class, and it matches the probe numbers. In both models the hottest region sits on the hand (palm/fingers), which is why changing the background does not hurt them. But the **baseline's attention is diffuse and visibly bleeds onto the background** (walls, floor, desk) — the same context leakage that let it score 0.522 with the hand hidden. The **SOTA crop model concentrates tightly on the hand**, because it never sees anything else. So the crop does not just match the baseline's accuracy; it makes the decision trustworthy.

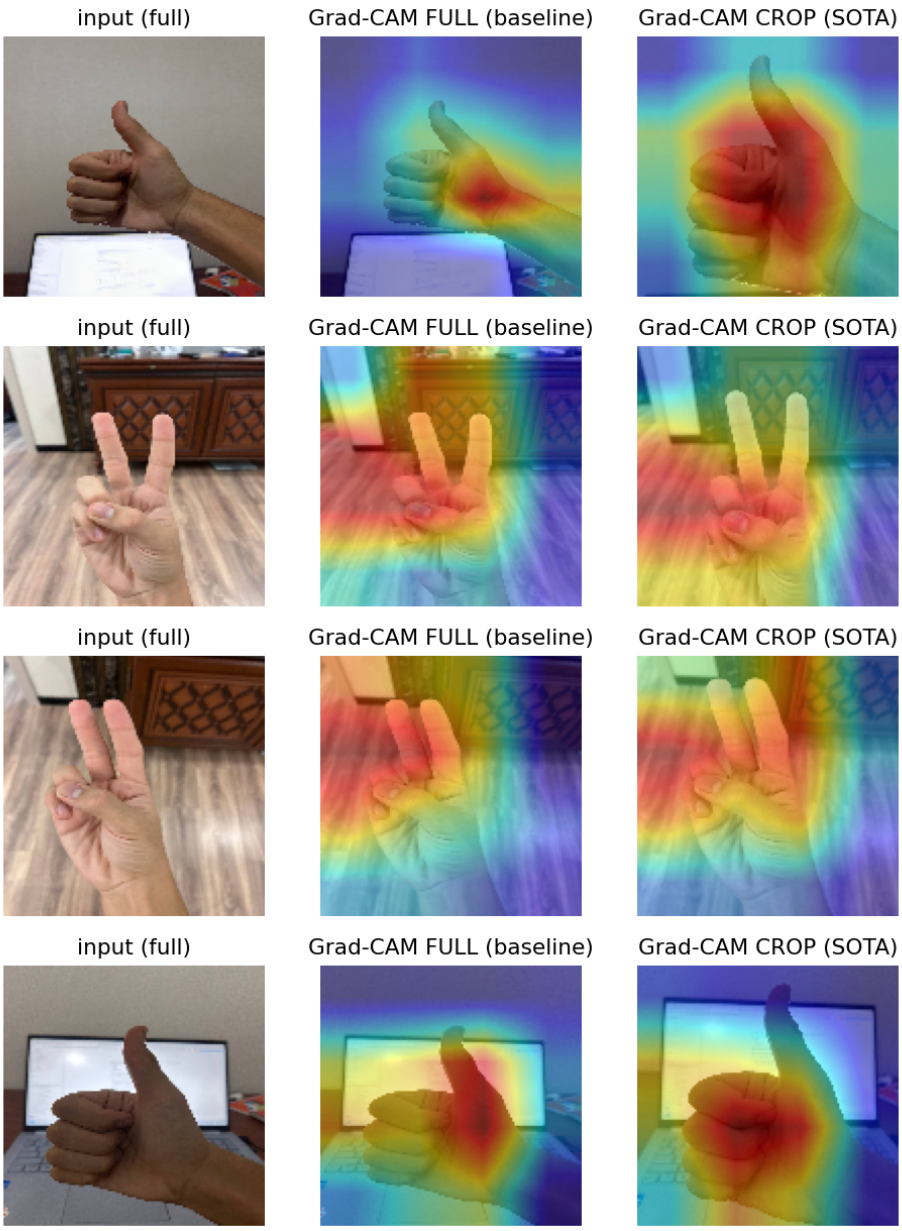



Figure 7. Grad-CAM: input (left), baseline on the full image (middle), SOTA on the crop (right). Baseline attention spills into the background; the crop model stays on the hand.

## 8. Ethics & limitations

**Consent & data.** The dataset contains only my own hand; no third party's biometric or personal data is involved, and the demo stores nothing. **Bias.** One person, one skin tone, one set of rooms — the model will not generalize to other people, cameras or lighting, and must not be deployed as-is. **Misuse.** Gesture systems can feed surveillance or always-on cameras; any real deployment needs explicit, revocable consent and on-device processing. **Disclosure.** No synthetic or generated media is presented as real; if generative augmentation were added it would be clearly labeled and watermarked. **Leakage honesty.** I explicitly flag that the clean accuracy is optimistic.

## 9. Reproducibility

Seed 42; image 128×128; stratified 70/15/15. Environment in requirements.txt; the hand_landmarker.task model auto-downloads. One command reproduces every number and figure: *python src/pipeline.py --data-dir ./ASSIGNMENT1CV --out results*. The Colab notebook runs the same code on a free GPU. The demo: *python webapp/app.py*. Measured latency: baseline 91 ms/img (11 FPS); SOTA end-to-end 135 ms/img (7 FPS) on CPU.

## 10. Conclusion

A2 upgrades the A1 gesture classifier with a genuine SOTA component (MediaPipe Hand Landmarker, 96% detection) and, more importantly, evaluates it honestly. On the leaky split both models are near ceiling, so accuracy is uninformative. The controlled probes are the real result: changing the background does not move either model, but hiding the hand halves the baseline's accuracy while leaving a clear background/context shortcut — a shortcut the MediaPipe crop removes by construction, as Grad-CAM confirms. The honest finding is therefore not 'SOTA is more accurate here' (the dataset is too easy and too leaky to show that) but 'SOTA makes the model use the right evidence', which is what matters for a new user or a new room. Future work: a multi-person dataset with session-aware splits (to remove the leakage), landmark-based classification, and generative augmentation (Step-Video-T2V / ArtAug) for rare conditions, clearly labeled.

### References

[1] F. Zhang et al., 'MediaPipe Hands: On-device Real-time Hand Tracking', 2020. [2] Google, MediaPipe Hand Landmarker documentation, ai.google.dev. [3] M. Sandler et al., 'MobileNetV2', CVPR 2018. [4] R. Selvaraju et al., 'Grad-CAM', ICCV 2017.